

Exponential Distribution

The exponential distribution is a continuous distribution that is sometimes used to model the time that elapses before an event occurs. Such a time is often called a waiting time.

The exponential distribution is sometimes used to model the lifetime of a component.

In addition, there is a close connection between the exponential distribution and the Poisson distribution.

The probability density function of the exponential distribution involves a parameter which is a positive quantity constant λ whose value determines the density function's location and shape.

The probability density function of the exponential distribution with parameter $\lambda > 0$ is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

If X is a random variable whose distribution is exponential with parameter $\lambda > 0$, then $X \sim \text{Exp}(\lambda)$.

The cumulative distribution function of the exponential distribution is easy to compute.

For $x \leq 0$, $F(x) = P(X \leq x) = 0$.

For $x > 0$, the cumulative distribution function is $F(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}$.

Mean and variance of exponential random variable

The mean and variance of an exponential random variable can be computed by using integration by parts.

If $X \sim \text{Exp}(\lambda)$, then $\mu_X = \frac{1}{\lambda}$ and $\sigma^2_X = \frac{1}{\lambda^2}$

Problems

If $X \sim \text{Exp}(2)$, then find μ_X ; σ^2_X ; and $P(X \leq 1)$.

$$\mu_X = \frac{1}{2} = 0.5; \sigma^2_X = \frac{1}{4} = 0.25; P(X \leq 1) = \int_0^1 e^{-2t} dt = 1 - e^{-2} = 0.865.$$

In a certain town, the duration of a shower is exponentially distributed with the mean 5 minutes. What is the probability that shower will last for

a) less than 10 minutes b) 10 minutes or more; c) between 10 and 12 minutes.

$\mu_X = \frac{1}{\lambda} = 5$ and x be the duration of the shower.

a) $P(\text{less than 10 minutes}) = P(x < 10) = 0.8646$.

b) $P(10 \text{ min or more}) = P(x \geq 10) = 0.1353$.

c) $P(\text{between 10 and 12}) = P(10 < x < 12) = 0.0446$.